# Independent Reconstruction and Exact Certificates for Ten Complete-Graph Incidence Configurations

Technical report accompanying the reproducible data package

10 October 2026

**Abstract**

All ten requested incidence-family configurations have been reconstructed and actually certified by the five unsigned recursive-line rules in the supplied manuscript. Three grids are transcriptions of its fully printed tables; the other seven are independently generated replacement witnesses. Every grid attains the cell bound, every selected pair is identified, and every distinct selected-edge class pair is certified orthogonal. This report specifies the mathematical rules, implementation, finite proof format, validation tests, search provenance and reproducibility commands. No unavailable original data or reported replay count is treated as a program result.

## 1 Outcome and the available source evidence

The input is the 10 October 2026 version of *Exact Second-Order Zarankiewicz Numbers for Complete-Graph Incidence Families* [1], together with the coauthor review note dated 9 October 2026 [2]. Sections 2.2–2.5 and Proposition 5.1 provide a precise certificate specification. Appendix A specifies the original CSV convention: blank for a singleton, a positive label occurring twice for a pair, and a full stop for a hole. Tables 8–10 in Appendices B–D print all cells at $n = 9, 8, 7$ respectively.

The seven larger appendices give counts and provenance, but not the endpoints of their repaired pairs. The supplied attachments contain neither the ten original CSV files nor the referenced checker. Counts alone do not determine those missing pairings. Therefore the seven larger files in this package are new witnesses, not claimed recovered originals. Their rule diagnostics and restriction counts need not equal those in Tables 1 and 7 of the manuscript.

For each $n \in \{7, 8, 9, 12, 13, 16, 17, 18, 20, 21\}$, the actual certificate establishes

$$R(G) = Z(n) = \left\lfloor \frac{n(n-1)(n+2)}{4} \right\rfloor, \qquad G \text{ satisfies } (\mathrm{RW3}^{+}).$$

Using the hierarchy and cell bound stated in the supplied manuscript, this gives $z_2 = z_{SL} = z_{RL} = Z(n)$ at these ten incidence dimensions. It does not settle any other order or any different seven-column table dimension.

## 2 Mathematical specification and five sound rules

Put $m = \binom{n}{2}$, $u_{ij} = x_i y_j$, and let the singleton set be exactly the incidence matrix of $K_n$. Simplicity means that every occupied cell belongs to exactly one selected singleton or selected pair. Write $\Omega$ for the occupied cells and $\delta(p, q) = 1$ precisely when the distinct cells $p, q$ form a selected pair; otherwise $\delta(p, q) = 0$.

For any alternative real SOS representation, let $v_p$ be the coefficient vector of cell $p$. Coefficient comparison yields

$$\begin{aligned} \|v_p\|^2 &= \mathbf{1}_\Omega(p), \\ \langle v_p, v_q \rangle &= \delta(p, q) && \text{for distinct cells on a common row or column,} \\ \langle v_p, v_q \rangle + \langle v_r, v_s \rangle &= \delta(p, q) + \delta(r, s) && \text{for the diagonals of a genuine rectangle.} \end{aligned}$$

Thus occupied vectors are unit vectors and hole vectors vanish. The following are the only allowed inference rules in the checker.

1. **Line.** Certify the prescribed value of each occupied common-line pair: identify its ends when $\delta = 1$, and declare them orthogonal when $\delta = 0$.

2. **Saturation.** Propagate an orthogonality fact to equivalent representatives. Equivalence is stored independently of orthogonality.

3. **Rectangle transfer.** If one diagonal has been certified at its own prescribed value, certify the other at its own prescribed value. Only distinct rows and distinct columns form a genuine rectangle.

4. **Complementary pair.** If both diagonals are selected pairs, identify the two ends of each. Their inner products sum to 2 and each is at most 1, so both equal 1.

5. **Zero companion.** If a diagonal contains a hole, its inner product and its prescribed value both vanish. Certify the opposite diagonal at its own prescribed value. In particular, a selected opposite pair is identified, not declared orthogonal.

Printed equal labels are never used as initial equivalences. No inference from an ungrounded signed odd cycle is admitted.

The terminal conditions are: both ends of every selected pair are identified; distinct selected edges remain in distinct equivalence classes; and all distinct selected-edge classes are pairwise orthogonal. They force $R(G)$ orthogonal unit vectors in every SOS representation, hence $\mathrm{SOS}(P_G) \geq R(G)$. The displayed SOS gives the reverse inequality. The singleton graph is $C_4$-free and has $2m = z(m, n)$, since a $C_4$-free matrix with row degrees $r_i$ satisfies

$$\sum_i \binom{r_i}{2} \leq \binom{n}{2} = m, \qquad \sum_i r_i \leq m + \binom{n}{2} = 2m.$$

Simplicity then gives $R(G) \leq \lfloor (mn + 2m)/2 \rfloor = Z(n)$. Attainment and the finite certificate suffice; no exhaustive exclusion is needed.

## 3 Exact implementation and independently checked finite proofs

The CSV audit checks shape, tokens, multiplicities, contiguous labels, singleton row positions and $C_4$-freeness. Blank fields are singleton cells, not missing values. Rows have no header and follow the lexicographic order of edges $\{a, b\}$, $1 \leq a < b \leq n$. Holes are excluded from the occupied equivalence structure.

The main checker uses Union-Find for identifications and a separate Boolean matrix for cell orthogonality. Under these rules every nontrivial equivalence class has at most two cells: every equality conclusion has prescribed value 1, which occurs only for a selected pair. Saturation propagates orthogonality over both representatives, then rectangle transfer supplies new orthogonality facts and new identifications. Initialization applies the line and complementary rules and, when enabled, the zero-companion rule. Iteration continues until neither store changes. All computations are exact Boolean or integer operations; no numerical tolerance or floating-point linear algebra enters acceptance.

For $N = mn$, one dense closure round costs $\mathcal{O}(N^2)$ operations and the working stores cost $\mathcal{O}(N^2)$ memory. A proof with $L$ rounds stores $\mathcal{O}(LN^2)$ bits before compression. This implementation prioritizes transparent exact verification over asymptotic scalability.

Each `logs/*_certificate.npz` is a finite synchronous proof: initial equality and orthogonality, the zero-companion stage, and the packed orthogonality matrix and newly identified pair indices at each subsequent round. Identifications use zero-based CSV-label indices. Boolean matrices use row-major order and NumPy's default big-endian bit packing.

`verify_certificate.py` does not invoke the closure routine. It independently enumerates row-pair and column-pair rectangle diagonals, checks initialization, verifies each round using only its legal premises, rejects missing or extra inferred facts, checks contradictions and finally tests all terminal conditions. This proof does not rely on trusting a claimed terminal count. File hashes bind the distributed proof and input to the package manifest.

`reference_checker.py` is a slow set-based implementation with a different update order and explicit rectangle loops. It agrees with the full cell-level equality and orthogonality stores on all three printed grids. Nine tests passed, including 80 seeded random small grids, each with the hole rule enabled and disabled; selected and unselected opposite diagonals; line and complementary identifications; malformed labels and a $C_4$; and rejection of a tampered finite certificate.

All ten saved proofs passed independent verification. All seven searched grids were also regenerated in a fresh directory and reproduced byte for byte. The three printed grids were re-extracted from the source PDF and likewise reproduced byte for byte.

## 4 Deterministic generators and sequential repair

For $q = 7, 11$, the parent vertices are two copies of $\mathbb{Z}_q$, with vertices $a$ and $a + q$. The explicitly implemented matchings are

$$A_a = \{\{a, a+q\}\} \cup \bigcup_{t=1}^{(q-1)/2} \{\{a+t, a-t\}, \{a+t+q, a-t+q\}\},$$
$$B_t = \{\{a, a+t+q\} : a \in \mathbb{Z}_q\}, \quad 1 \le t < q,$$

where arithmetic within either copy is modulo $q$. The order of inner labels is the anchor, then the two reflection edges at $t = 1, 2, \ldots$; the $B_t$ edges are ordered by $a = 0, 1, \ldots, q-1$.

For $q = 9$, the implemented parent uses the cyclic factorization on $\mathbb{Z}_{17} \cup \{17\}$:

$$F_a = \{\{a, 17\}\} \cup \{\{a+t, a-t\} : 1 \le t \le 8\}, \quad a \in \mathbb{Z}_{17}.$$

Its inner labels are in the displayed order. For each factor, each anchor edge with inner label $b$ and endpoints $u, v$, and each $1 \le x \le (q-1)/2$, the rows indexed by inner labels $b + x$ and $b - x$ are paired on columns $u, v$ along both complementary diagonals. This uses each non-incidence cell exactly once.

The generated $q = 7$ and $q = 11$ parents pass the exact closure. The cyclic $q = 9$ parent fails the ordinary criterion and leaves exactly 7,344 selected-edge pairs uncertified, reproducing the manuscript's parent diagnostic. A preliminary bipyramidal composite-$q$ attempt instead left 163,755 pairs uncertified; its failed input and log are retained separately. It is not the cyclic parent used for the successful $n = 16, 17$ witnesses.

Repair keeps the vertices $0, \ldots, n-1$, deletes all other vertex columns and incident edge rows, retains complete surviving pairs, and collects the orphaned cells. The cells are paired using seed 20261010. The search first minimizes the number of unidentified pairs, then the number of uncertified selected-edge class pairs. It switches endpoints among two to four pairs, with a fresh random proposal every twentieth attempted iteration. The targeted variant selects an endpoint involved in a current unresolved relation and permits neutral moves. The earlier basic variant, used at $n = 13, 12$, uses strict improvements and untargeted switches.

The order was $7, 8, 9$ (printed), then $13, 12, 16, 17, 18, 20, 21$ (searched). The $n = 12$ repair uses the newly found $n = 13$ parent. No surviving inherited pair was deliberately broken in the seven successful replacement searches. The $n = 18$ deletion keeps 552 pairs and frees 1,344 cells; these differ from the manuscript's 560 and 1,328 because the independently specified parent and deleted set are not asserted to be its unpublished choice.

## 5 Actual verification results

| $n$ | $m$ | $\lvert E_1\rvert$ | $\lvert E_2\rvert$ | $H$ | $R = Z(n)$ | orthogonal class pairs |
|---|---|---|---|---|---|---|
| 7 | 21 | 42 | 52 | 1 | 94 | 4,371 |
| 8 | 28 | 56 | 84 | 0 | 140 | 9,730 |
| 9 | 36 | 72 | 126 | 0 | 198 | 19,503 |
| 12 | 66 | 132 | 330 | 0 | 462 | 106,491 |
| 13 | 78 | 156 | 429 | 0 | 585 | 170,820 |
| 16 | 120 | 240 | 840 | 0 | 1080 | 582,660 |
| 17 | 136 | 272 | 1020 | 0 | 1292 | 833,986 |
| 18 | 153 | 306 | 1224 | 0 | 1530 | 1,169,685 |
| 20 | 190 | 380 | 1710 | 0 | 2090 | 2,183,005 |
| 21 | 210 | 420 | 1995 | 0 | 2415 | 2,914,905 |

Every row has status `VERIFIED`, exactly $R$ occupied equivalence classes, exactly $|E_2|$ identifications, zero uncertified class pairs, and no equality/orthogonality conflict. Counts are computed from the CSV inputs, not inserted from a source table.

| $n$ | line | complementary | zero | transfer | rounds | seconds |
|---|---|---|---|---|---|---|
| 7 | 11 | 6 | 0 | 35 | 41 | 0.0083 |
| 8 | 26 | 24 | 0 | 34 | 39 | 0.0135 |
| 9 | 51 | 72 | 0 | 3 | 12 | 0.0090 |
| 12 | 17 | 168 | 0 | 145 | 28 | 0.0835 |
| 13 | 8 | 312 | 0 | 109 | 16 | 0.0702 |
| 16 | 17 | 528 | 0 | 295 | 18 | 0.2729 |
| 17 | 14 | 816 | 0 | 190 | 14 | 0.3367 |
| 18 | 34 | 552 | 0 | 638 | 34 | 1.0460 |
| 20 | 22 | 1200 | 0 | 488 | 16 | 1.0734 |
| 21 | 8 | 1680 | 0 | 307 | 14 | 1.4788 |

Rule counts here count selected-pair identifications, in the stated initialization order. They reproduce the manuscript's counts at $n = 7, 8, 9$. The seven replacement witnesses have their own diagnostics. Closure times exclude compression and independent proof verification and are observations from this run, not performance guarantees. The JSON logs additionally give new unordered cell orthogonality facts by rule, including saturation.

At $n = 7$, the hole is the cell in row $\{2, 7\}$ and column 4. There are 120 genuine rectangles containing it. With the rule disabled, every selected pair is still identified but only 4,350 of 4,371 class pairs are certified. The quotient transfer graph has 322 components: 16 are ungrounded, with sizes twelve 1's, three 2's and one 3. Enabling the hole rule grounds those 16 components and certifies the remaining 21 variables. These diagnostics were recomputed, not copied from Appendix D.

# 6 Search records, replay and scope

| $n$ | inherited pairs | free cells | success attempt (one-based) | search seconds |
|---|---|---|---|---|
| 12 | 236 | 188 | 245 | 21.45 |
| 13 | 312 | 234 | 68 | 5.44 |
| 16 | 528 | 624 | 11 | 2.93 |
| 17 | 816 | 408 | 8 | 2.67 |
| 18 | 552 | 1344 | 26 | 24.70 |
| 20 | 1200 | 1020 | 16 | 15.85 |
| 21 | 1680 | 630 | 11 | 15.77 |

The detailed JSONL files preserve seed, parameters, each evaluated score and measured elapsed time. A skipped targeted proposal need not produce a score record. The attempt index records the actual loop index. Original searches had a 180-second budget and stopped upon acceptance. Deterministic regeneration disables that time cutoff while retaining the recorded attempt cap, so differing processor speeds do not alter the intended search prefix.

From the extracted package root, the essential replay commands are

```
python -m pip install -r ancillary/requirements.txt
python ancillary/validate_all.py
python ancillary/test_checker.py
python ancillary/regenerate_all.py ../fresh_reconstruction
```

For an individual case, for example $n = 7$:

```
python ancillary/certificate_checker.py ancillary/217.csv
python ancillary/verify_certificate.py ancillary/217.csv \
    logs/217_certificate.npz
```

The environment used was Python 3.12.14 with NumPy 2.3.5. The core checker, proof verifier and reconstruction require only Python and NumPy. Poppler is optional for repeating PDF extraction; a LaTeX installation is optional for rebuilding this report.

The package contains all ten CSV grids, the checker and independent proof verifier, the slow reference implementation, tests, generator and regeneration script, ten finite NPZ certificates, actual run logs, the verification summary, provenance, requirements and SHA-256 manifest. The CSV checksums, per-sheet commands and expected outputs are indexed in `ancillary/README.md`. This report's source and compiled PDF are included.

These finite witnesses establish the ten stated incidence dimensions under the supplied mathematical framework. They do not reconstruct the missing original larger sheets, reproduce their particular local-search histories, verify the eight different seven-column entries $9 \le m \le 16$, or establish the bounded-gap conjecture in general. The seven replacement witnesses require corresponding changes to the manuscript's construction provenance and replay tables. In particular, source Table 1 is not a checksum or a complete construction recipe. For a public paper, this actual replacement package should be deposited at an immutable archive and cited as a new version; an archive location has not been invented here.

# References


[1] Y. Chen, J. Löfberg and L. Qi, *Exact Second-Order Zarankiewicz Numbers for Complete-Graph Incidence Families*, supplied manuscript, 10 October 2026, file `more(5).pdf`.

[2] *Changes needed in more_8.tex*, notes for the coauthors, 9 October 2026, supplied file `more_8_review_note.pdf`.